
Retrieval-Guided Data Assimilation for Observation-Aware Video World Models

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Abstract

Modern image-to-video generators can be read as video world models: given a current visual state and a task description, they roll out a latent trajectory that represents how the scene may evolve. Under this view, failures of motion smoothness and physics consistency are not merely visual artifacts, but world-model state-estimation errors: the model’s latent forecast drifts away from plausible real dynamics. This paper formulates retrieval-guided Video-as-Prompt inference as a data-assimilation process for observation-aware video world models. Given an input image and prompt, a retrieval memory supplies one semantically related real-video trajectory from OpenVid-style memory. This trajectory is used as the reference-video observation in the existing in-context DiT pathway, where generated latent forecast tokens interact with retrieved observation tokens. The forecast-side output is interpreted as the analysis state used for noise prediction, while forecast-to-observation attention weights provide a compatibility signal for motion smoothness and physical fidelity. As preliminary evidence, we report a Moving MNIST latent-ETKF study and a six-case qualitative pilot comparing Wan2.1-14B image-to-video baseline inference with retrieved-video inference. The pilot suggests that retrieved observations can increase scene-level motion, while also revealing blur, framing drift, and runtime costs that motivate future work.

1 Introduction

A useful video world model must do more than produce visually plausible frames. Following the “Trinity of Consistency” view of general world models, it should satisfy three coupled requirements: modal consistency, which aligns language, images, and video with the same semantic state; spatial consistency, which preserves the geometry, layout, and identity of the world; and temporal consistency, which makes the state evolve causally over time [23]. This trinity is a sharper starting point for our problem than generic video quality. If a generated video violates spatial consistency, objects can drift, pop, or reappear with different geometry. If it violates temporal consistency, motion becomes jittery or physically implausible. If it violates modal consistency, the video may be smooth but no longer follow the prompt or anchor image.

Recent video world-model work has made the spatial part of this trinity especially visible. World-consistent diffusion models introduce explicit 3D coordinate supervision to make generated frames geometrically coherent [27]. GEN3C uses a 3D cache to avoid object popping and improve world-consistent camera-controlled generation [15]. Long-term spatial memory methods explicitly store

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and retrieve 3D scene memory because limited temporal context causes forgetting during revisits [25], while compressed memory approaches such as WorldPack improve spatial consistency by packing and retrieving past trajectories [13]. These works point to a common diagnosis: current video generators are not yet reliable world models because their latent state is weakly anchored to persistent spatial evidence and real dynamical observations.

However, modern text/image-to-video models typically roll out the world state without real-time observations. The model receives an anchor image and a text prompt, then predicts a video consistent with that initial state and intent [20]. This interface gives a strong modal condition, but once generation begins, spatial and temporal consistency are left to the model’s internal prior. The model must simulate dynamics from limited conditioning, without access to external observations of how similar scenes usually evolve. The result can be temporally irregular motion, weak object persistence, and physically implausible state transitions.

Inspired by classical data assimilation (DA), where a model forecast is corrected by external observations to produce an analysis state, we transfer the same logic into the DiT denoiser of an image-to-video world model. The noisy generated latent tokens are the forecast state of the world model, the single retrieved real-video token sequence is an external observation of plausible spatial-temporal evolution, and cross attention between them is the learned analysis operator that updates the representation used for noise prediction. In this framing, spatial consistency, motion smoothness, and physics fidelity become state-estimation properties: generated tokens should remain compatible with real observed dynamics in the retrieval memory.

Our current contributions are:

1. We frame retrieval-guided image-to-video generation as an observation-aware video world-model problem grounded in modal, spatial, and temporal consistency.
2. We use a single retrieved real-video trajectory as observation tokens for the forecast latent state, while relying on the existing in-context DiT pathway rather than introducing a new attention mechanism.
3. We connect the resulting forecast-to-observation token interaction to the classical forecast–observation–analysis structure of data assimilation and provide preliminary controlled and qualitative evidence.

2 Preliminaries and Preliminary Study

2.1 Classical DA Framing: ETKF

Classical DA provides the forecast–observation–analysis template used in this paper. The preliminary experiment instantiates this template with an ensemble transform Kalman filter (ETKF) [2]. Let $\{x_i^f\}_{i=1}^M$ be an M -member forecast ensemble, with ensemble mean $\bar{x}^f$ and anomaly matrix

$$X'^f = [x_1^f - \bar{x}^f \quad \dots \quad x_M^f - \bar{x}^f]. \quad (1)$$

Given an observation y , an observation operator H , and observation covariance R , the corresponding forecast observations are $y_i^f = H(x_i^f)$. Their mean and anomaly matrix are $\bar{y}^f$ and

$$Y'^f = [y_1^f - \bar{y}^f \quad \dots \quad y_M^f - \bar{y}^f]. \quad (2)$$

ETKF performs the analysis update in ensemble space:

$$S = (M - 1)I + (Y'^f)^\top R^{-1} Y'^f, \quad (3)$$

$$w^a = S^{-1} (Y'^f)^\top R^{-1} (y - \bar{y}^f), \quad (4)$$

$$T^a = [(M - 1)S^{-1}]^{1/2}. \quad (5)$$

The analysis mean and analysis anomalies are then

$$\bar{x}^a = \bar{x}^f + X'^f w^a, \quad X'^a = X'^f T^a. \quad (6)$$

Thus the analysis ensemble is centered at a corrected state while preserving an updated ensemble spread. In the Moving MNIST preliminary study, x_i is not an image directly; it is the latent recurrent state of the video predictor. This ETKF view is the controlled DA analogue of the retrieval-guided video setting: a forecast state is corrected by an observation trajectory before the next rollout step.

2.2 Preliminary Moving MNIST Study

Before evaluating retrieval-guided motion and physics consistency with a full image-to-video backbone, we include a controlled Moving MNIST study [19] to isolate the DA interpretation itself. This experiment should be read as a partial result rather than as the final retrieval-guided Video-as-Prompt benchmark: it does not test OpenVid retrieval or Wan generation. Instead, it asks whether a learned dynamical predictor becomes smoother and more faithful when sparse observations are assimilated into the latent state of the same host predictor.

The task uses the standard 10-to-10 Moving MNIST prediction setup. The first ten frames form the context. The open-loop method is a lightweight latent recurrent host, `OpenLoopVideoPredictor`, built from a residual convolutional encoder-decoder and a ConvLSTM-style transition module [18]. After encoding the context frames, the host rolls out future frames autoregressively: the recurrent state predicts a residual update to the previous latent frame, and the decoder maps each predicted latent back to image space. No future observations are used in this open-loop row.

The open-loop+DA method uses the same host checkpoint and changes only the test-time state update. We form an ensemble by perturbing the host latent recurrent state (z_t, h_t, c_t) , propagate all ensemble members through the same forecast step, decode the forecast ensemble into observation space, and apply the latent ETKF update above whenever an observation is available. In the reported DA condition, the observation operator reveals the full held-out future frame every two future steps and adds Gaussian observation noise with standard deviation 0.02, yielding five analysis updates per sequence. The analysis mean is decoded as the corrected prediction and is also fed forward as the state from which the next forecast starts.

We report root mean squared error (RMSE) and pattern correlation coefficient (PCC). For predicted frames $\hat{x}$ and target frames x , RMSE is

$$\text{RMSE}(\hat{x}, x) = \sqrt{\frac{1}{|\Omega|} \sum_{u \in \Omega} (\hat{x}_u - x_u)^2}, \quad (7)$$

where Ω indexes all evaluated pixels across the predicted sequence. Lower RMSE indicates smaller pixel-level error. PCC measures frame-wise structural agreement after removing each frame mean:

$$\text{PCC}(\hat{x}, x) = \frac{1}{T} \sum_{t=1}^T \frac{\langle \hat{x}_t - \bar{\hat{x}}, x_t - \bar{x} \rangle}{\|\hat{x}_t - \bar{\hat{x}}\|_2 \|x_t - \bar{x}\|_2}. \quad (8)$$

Higher PCC indicates better agreement in the predicted spatial pattern.

Table 1: Preliminary Moving MNIST latent-DA result. All DA rows use the same host predictor as the open-loop row, with sparse full-frame observations every two future steps. Lower RMSE, higher PCC, and lower runtime ratio are better.

Method	RMSE ↓	PCC ↑	Observed steps	Runtime ratio
Open-loop host	0.1495	0.7593	0	1.00×
Host + latent ETKF (8 members)	0.1410	0.7811	5	1.85×
Host + latent ETKF (100 members)	0.1104	0.8375	5	11.97×
Host + latent ETKF (500 members)	0.0898	0.8915	5	40.36×

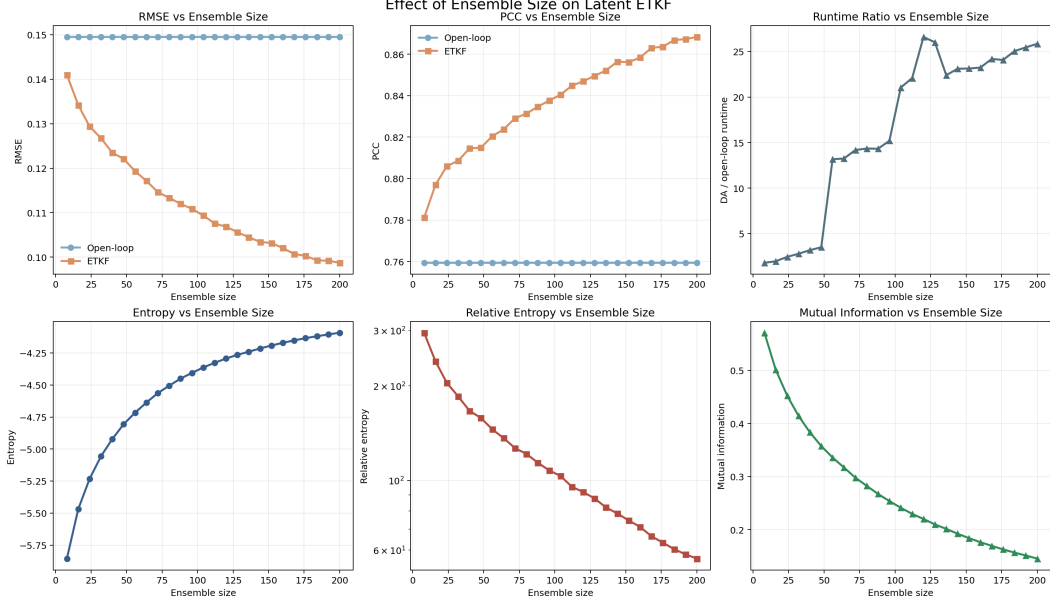


Figure 1: Previously run Moving MNIST latent-ETKF ensemble-size sweep. The RMSE panel shows that larger ensembles improve the ETKF analysis quality, while the runtime panel shows the rapidly increasing cost of explicit ensemble assimilation.

Table 1 shows that latent ETKF improves the same host predictor as the ensemble size grows. The 8-member update gives a small but consistent gain, reducing RMSE from 0.1495 to 0.1410 and improving PCC from 0.7593 to 0.7811. The 100-member and 500-member runs strengthen the correction further, reaching RMSE 0.1104/PCC 0.8375 and RMSE 0.0898/PCC 0.8915, respectively. Figure 1 shows the same trend in the previous ensemble-size sweep up to 200 members. The result is useful for this paper because it demonstrates the same conceptual decomposition used by the retrieval-guided inference pipeline for motion quality: a forecast state, an observation operator, an innovation, and an analysis update.

This controlled result also clarifies the expected tradeoff. Stronger explicit ETKF correction requires larger ensembles, and runtime grows from $1.85\times$ at 8 members to $11.97\times$ at 100 members and $40.36\times$ at 500 members. This runtime scaling motivates using the existing in-context attention pathway to approximate the ETKF analysis process inside the generator rather than explicitly sampling and updating hundreds of trajectories. In the video setting, forecast tokens can play the role of the forecast state, retrieved real-video tokens can play the role of observations, and in-context attention can approximate the analysis update in a single denoising pathway. Thus, the preliminary study motivates the Video-as-Prompt inference version of the method: the retrieved video should act as the observation trajectory inside the DiT in-context pathway for smoother, more physically plausible motion.

3 Method

3.1 Overview

The method uses Video-as-Prompt-style inference [1] as the host image-to-video system and interprets its existing in-context pathway as an attention-based approximation to a DA analysis update. The base image-to-video branch is the forecast stream. The retrieved reference-video branch is the observation stream. The existing in-context DiT pathway allows these streams to interact without explicitly sampling a large ETKF ensemble. The pipeline is organized into four steps:

1. retrieve one real trajectory from the shared LanguageBind world memory;
2. encode the selected trajectory as observation tokens O ;
3. run Video-as-Prompt inference with noisy forecast tokens H^f and observation tokens H^o ;

4. interpret the forecast-side in-context output as analysis tokens before noise prediction.

Figure 2 summarizes the retrieval and DiT inference architecture.

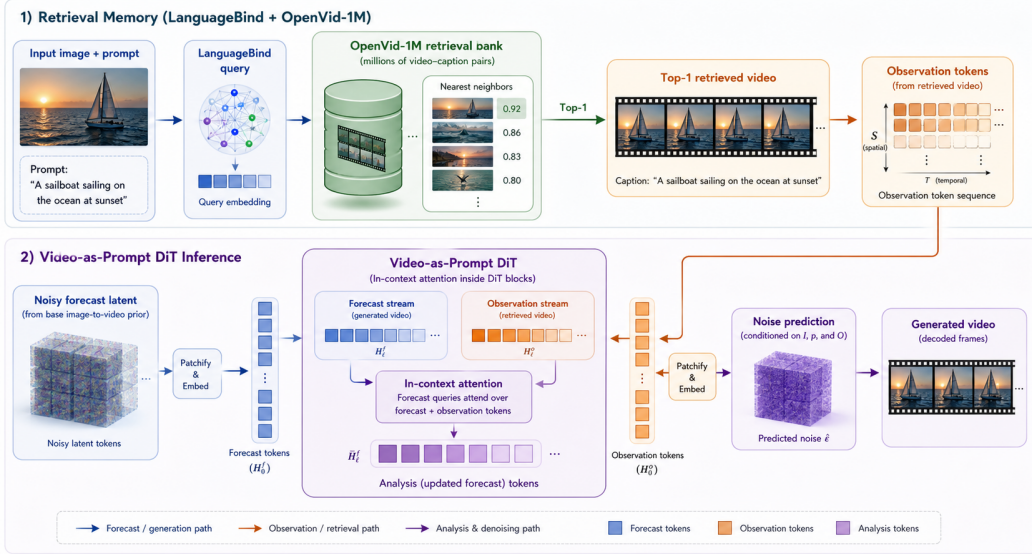


Figure 2: Architecture of the retrieval-guided Video-as-Prompt inference pipeline. The input image and text prompt form a LanguageBind query over an OpenVid-1M retrieval bank. The top-1 retrieved real clip is encoded as observation tokens, while the base image-to-video branch supplies noisy forecast tokens. Inside the Video-as-Prompt DiT denoiser, forecast tokens attend to retrieved observation tokens; the forecast-side analysis tokens are then used for noise prediction and video generation.

3.2 Retrieval Memory and Shared Embedding Space

We assume a retrieval bank

$$\mathcal{R} = \{(r_i, c_i)\}_{i=1}^N, \quad (9)$$

where r_i is a real video trajectory and c_i is its caption. A LanguageBind encoder [30] maps text, image, and video inputs into a shared embedding space. Given a prompt p and an anchor image I , we build the retrieval query

$$q = Q_{\text{LB}}(p, I), \quad (10)$$

and retrieve the single nearest clip

$$(r^*, c^*) = \arg \max_{(r_i, c_i) \in \mathcal{R}} s(q, r_i). \quad (11)$$

This retrieval memory defines the external observation source. The retrieved clip is not a label and not a target future. It is an empirical trajectory from world memory: one example of how a semantically related scene evolves in real video.

3.3 Video World Model Setting

Let an image-to-video world-model input be an anchor observation and a task descriptor

$$I \in \mathcal{I}, \quad p \in \mathcal{P}. \quad (12)$$

The model samples a latent trajectory and decodes it as a video

$$\hat{x} = (\hat{x}_1, \dots, \hat{x}_T) \quad (13)$$

from a noisy latent state z_t^p , the prompt p , and the anchor image I . Here, z_t^p is the native DiT latent state at denoising time t , and the superscript p denotes the prompt/image-conditioned prior rollout before any retrieved-video observation is used. We define the corresponding DA forecast state as

$$z_t^f \equiv z_t^p. \quad (14)$$

The superscript f denotes the forecast state: before assimilation, the world-model forecast is exactly the base DiT latent generated from (I, p) . The retrieved real-video clip is an observation of plausible motion and physical evolution for a semantically related scene. The goal is not to reconstruct a specific target clip, but to produce an analysis trajectory z_t^a whose decoded video is more stable, smoother, and more physically faithful than the unassimilated forecast.

In world-model terms, retrieval-guided inference is a latent state-estimation problem with an external memory. The base DiT defines a transition forecast from (I, p) to a noisy latent trajectory. Retrieval provides a single empirical observation trajectory O showing how a similar state evolves. The desired update is an analysis operator

$$z_t^a = \mathcal{A}_\theta(z_t^f, O, p, I), \quad (15)$$

where z_t^a is the analysis latent state after incorporating the retrieved observation at denoising step t ; $\mathcal{A}_\theta$ is the analysis operator induced by the Video-as-Prompt in-context DiT pathway with parameters θ ; z_t^f is the forecast latent state from the base image-to-video DiT before observation use; O is the single retrieved real-video observation-token sequence; p is the text prompt; and I is the input anchor image. The analysis state should preserve the semantic intent of the prompt while improving temporal coherence and physical plausibility.

3.4 Observation Tokens

The retrieved video trajectory is converted into one token sequence rather than a single summary vector or a multi-sample set. Let E_{vid} denote the frozen video preprocessing, VAE encoding, and patch embedding path that maps the retrieved clip into DiT-compatible spatio-temporal tokens. For the most similar retrieved clip, we write the observation token as:

$$O = E_{\text{vid}}(r^*, c^*), \quad (16)$$

where the caption c^* supplies text-side observation context. We do not concatenate multiple retrieved clips or use retrieval weights as attention biases in this first version. The key point is that the retrieved video enters the DiT as observation data for the world state, rather than as an external score or auxiliary label.

3.5 Forecast and Observation Streams

At denoising step t , the base DiT receives a noisy latent video z_t^f together with prompt and image conditioning. After patch embedding, this produces forecast tokens

$$H_0^f = P_{\text{patch}}(z_t^f, I, p). \quad (17)$$

The retrieved video branch receives the observation tokens O and optional observation text/image embeddings, giving

$$H_0^o = P_{\text{obs}}(O). \quad (18)$$

The forecast stream is the internal world-model rollout. The observation stream is a real-video branch. Both streams are propagated through the DiT, but only the forecast-side analysis output is used to predict the noise for the generated video.

3.6 Data-Assimilation View of In-Context Attention

For a selected DiT block ℓ , let H_ℓ^f denote forecast tokens and H_ℓ^o denote observation tokens. In the Video-as-Prompt-style in-context pathway, the generated-video branch and reference-video branch can be written with separate QKV projections:

$$Q_\ell^f, K_\ell^f, V_\ell^f = W_{qkv}^f H_\ell^f, \quad Q_\ell^o, K_\ell^o, V_\ell^o = W_{qkv}^o H_\ell^o. \quad (19)$$

The in-context full-attention form is

$$\begin{bmatrix} \bar{H}_\ell^f \\ \bar{H}_\ell^o \end{bmatrix} = \text{Attn} \left(\begin{bmatrix} Q_\ell^f \\ Q_\ell^o \end{bmatrix}, \begin{bmatrix} K_\ell^f \\ K_\ell^o \end{bmatrix}, \begin{bmatrix} V_\ell^f \\ V_\ell^o \end{bmatrix} \right), \quad (20)$$

where $\text{Attn}(Q, K, V) = \text{softmax}(QK^\top/\sqrt{d})V$. For a lighter cross-attention implementation, the forecast stream can query both streams while the observation stream is left fixed:

$$\bar{H}_\ell^f = \text{Attn}\left(Q_\ell^f, \begin{bmatrix} K_\ell^f \\ K_\ell^o \end{bmatrix}, \begin{bmatrix} V_\ell^f \\ V_\ell^o \end{bmatrix}\right). \quad (21)$$

In both cases, the forecast-side output $\bar{H}_\ell^f$ is the analysis representation. The cross terms between Q_ℓ^f and K_ℓ^o measure which observation tokens are relevant to the current generated latent forecast. We interpret these weights as an in-context score for motion smoothness and physical compatibility.

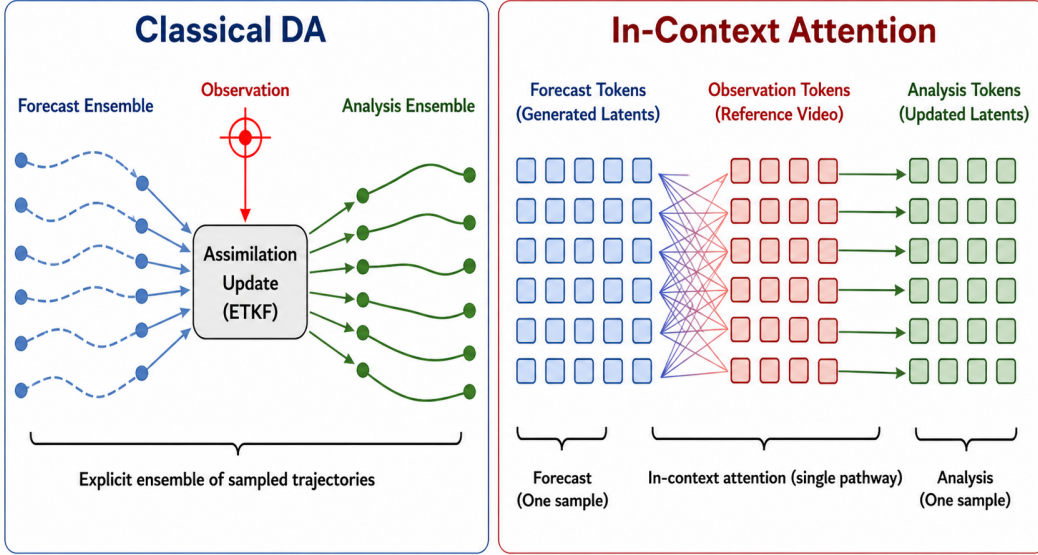


Figure 3: Conceptual comparison between explicit classical DA and in-context attention. Classical DA, shown on the left, performs an explicit ensemble update: many sampled forecast trajectories are corrected by an observation to produce an analysis ensemble. In-context attention, shown on the right, replaces explicit ensemble propagation with a single token pathway: generated forecast tokens attend to retrieved observation tokens and produce updated analysis tokens inside the DiT.

3.7 Noise Prediction with Observation Tokens

After the in-context interaction, the DiT predicts the noise using the updated forecast stream:

$$\hat{\epsilon} = \epsilon_\theta\left(z_t^f, t, p, I, O\right). \quad (22)$$

The observation tokens are therefore part of the DiT world-model input. They do not produce a separate output video; they influence the generated trajectory by modifying the forecast stream during attention.

4 Experiment

Retrieval bank and algorithm. OpenVid-1M serves as the external observation memory [12]. We retrieve the top-1 real clip by ranking cached LanguageBind caption/video embeddings against the input prompt and anchor image.

Inference with retrieved video. We compare a Wan2.1-14B baseline, conditioned only on the input image and prompt, with our Video-as-Prompt path [1], which uses the same image as the VAP reference image and the retrieved clip as the VAP reference video. Both paths generate 17 frames; the comparison tests whether the retrieved observation improves motion smoothness, physical consistency, and scene-level dynamics relative to the baseline.

5 Qualitative Experiment

Qualitative temporal samples: cases 1-2

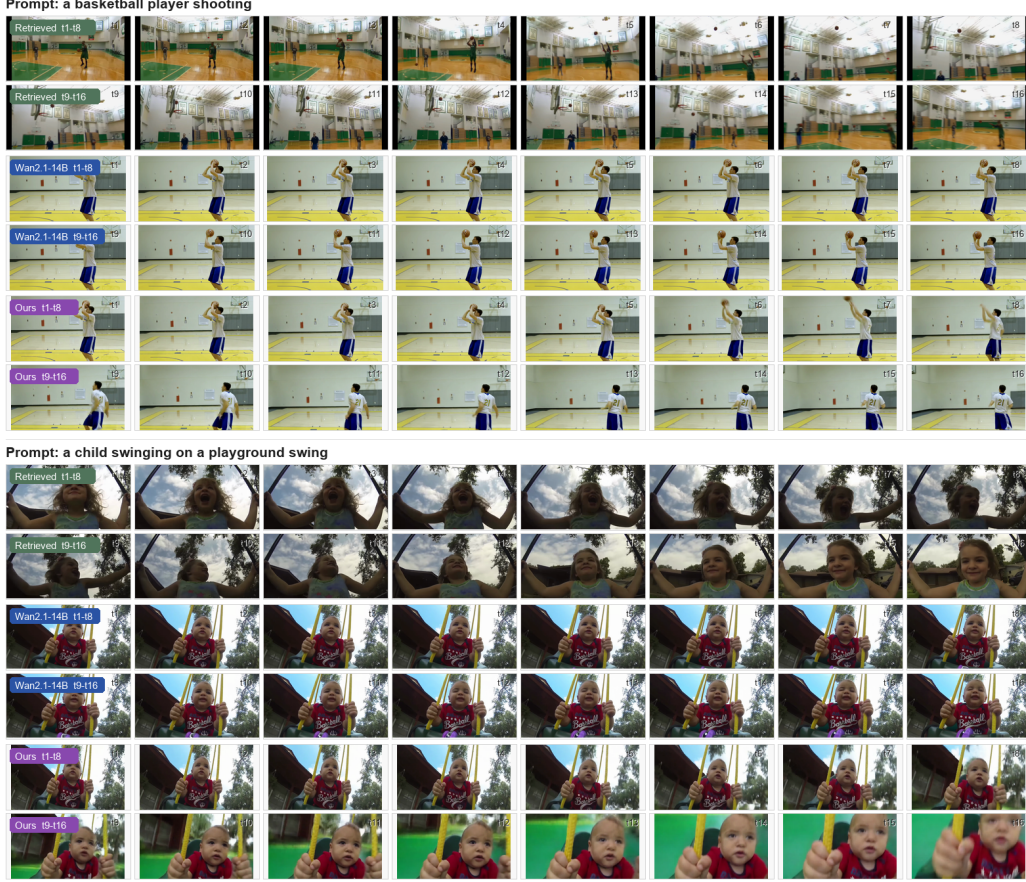


Figure 4: Temporal-sample qualitative comparison for cases 1–2. The exact text prompt is printed above each case. For each retrieved, Wan2.1-14B, or ours sequence, 16 temporally ordered samples are split into two rows of eight frames, t_1-t_8 and t_9-t_{16} .

Protocol. We run a first qualitative pilot on six manually supplied image–video pairs. Each folder name is converted into the prompt by replacing underscores with spaces. The figure panels serve as part of the experiment description: each case prints its exact prompt above the retrieved reference video, Wan2.1-14B baseline, and our retrieved-video generation. For each case, the Wan2.1-14B baseline receives only the input image and text prompt. The retrieval-guided run receives the same image and prompt, plus the corresponding reference video as the retrieved observation supplied to Video-as-Prompt inference. All runs use a single seed and generate 17 frames. We use Wan2.1-14B for the baseline test, matching the base family used by the retrieved-video Video-as-Prompt path. Therefore this is a diagnostic pilot rather than a full benchmark. Figures 4–6 show 16 temporally ordered samples spanning each available video. For each retrieved, Wan2.1-14B, or ours sequence, the samples are arranged as two rows of eight frames: t_1-t_8 and t_9-t_{16} . These labels denote normalized temporal sample positions, not consecutive frame indices. The retrieved rows are sampled from the full reference video used by Video-as-Prompt, while the Wan2.1-14B and ours rows are sampled across the generated 17-frame clips.

Qualitative temporal samples: cases 3-4

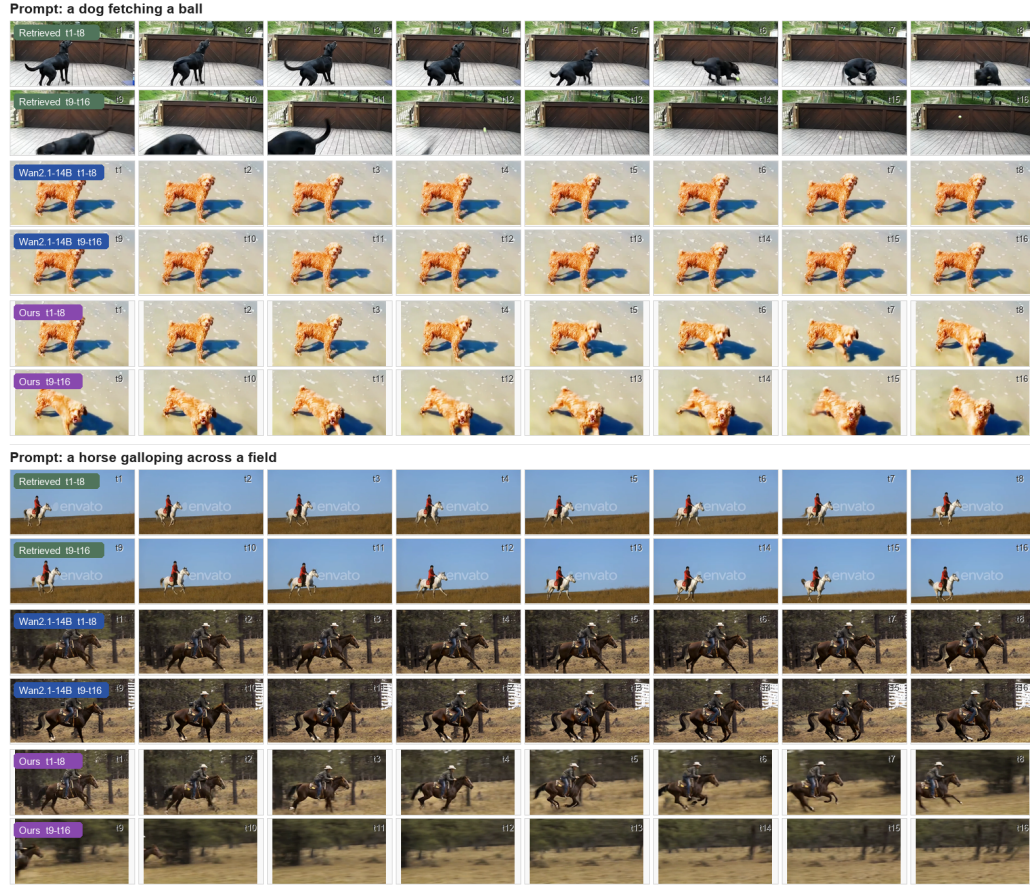


Figure 5: Temporal-sample qualitative comparison for cases 3–4. The retrieved rows show the real-video observation supplied to Video-as-Prompt, and the generated rows show Wan2.1-14B baseline inference and our retrieved-video inference at the same 16 normalized temporal positions.

Qualitative temporal samples: cases 5-6

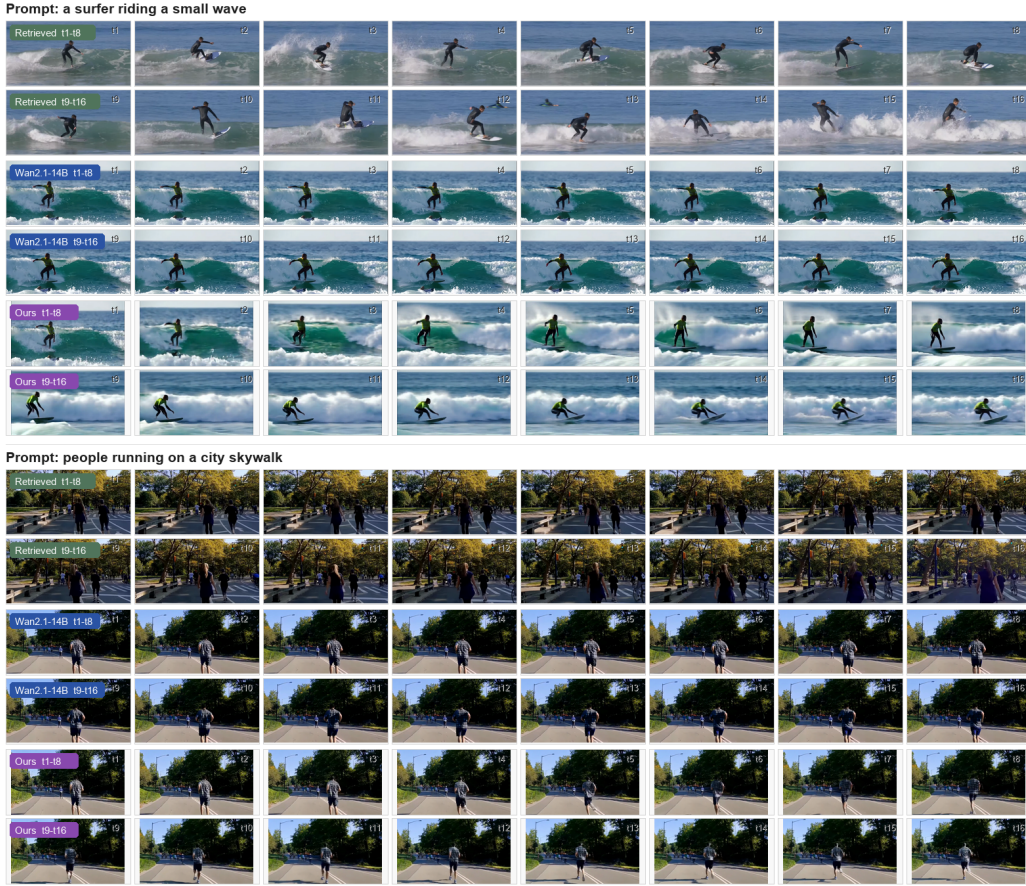


Figure 6: Temporal-sample qualitative comparison for cases 5–6. The two-row temporal layout makes scene-level motion visible while keeping each frame large enough to inspect.

Table 2: Qualitative observations from the six-case pilot. These are visual notes from side-by-side videos, not benchmark scores.

Case	Wan2.1-14B baseline	Ours + retrieved video observation
Basketball	Preserves the court and player cleanly, but most motion is local and the background remains nearly fixed.	Adds a clearer shooting-and-follow-through trajectory, with stronger body displacement; some appearance and layout drift appear.
Swing	Keeps the child’s face, ropes, and background stable, with only modest forward motion.	Produces stronger swing-like camera motion and depth change, but becomes close-up, blurry, and less identity-preserving.
Dog	The dog remains mostly stationary, so the result is visually stable but weakly dynamic.	Introduces body turning and approach motion in the water, with stronger temporal change but softer details.
Horse	Maintains horse and rider detail with a plausible gallop pose, but motion is relatively conservative.	Injects stronger lateral tracking and scene motion; the motion is more energetic but overshoots the frame by the late step.
Surfing	Keeps the wave scene readable and stable, with moderate motion.	Increases board, wave, and camera motion, making the clip more dynamic while adding blur and subject drift.
Running	Moves the foreground runner while much of the road and crowd remain static.	Produces stronger forward running and global scene motion; the road and distant runners evolve more visibly, with mild blur.

Analysis. Across the six cases, the retrieved-video observation most clearly helps when the prompt requires scene-level motion rather than only local texture change. Running, surfing, dog fetching, and basketball show the intended behavior: the observation path pushes the generated clip toward a more dynamic trajectory. This is consistent with the DA interpretation, because the retrieved video acts as an external motion observation that changes the forecast trajectory rather than only decorating the first frame. The negative side is equally important. The retrieved-video path sometimes sacrifices spatial fidelity, especially for the swing and horse examples, where the model introduces strong motion but loses framing or sharpness. The average runtime in this pilot is also higher: about 244 seconds for Wan2.1-14B baseline inference and about 810 seconds for the retrieved-video path, roughly a $3.3\times$ increase in this implementation. The result should therefore be read as a proof-of-concept signal for observation-aware dynamics, not as a final performance claim.

6 Discussion

The main benefit of the revised formulation is cleanliness: one video world-model prior, one retrieval memory, and one Video-as-Prompt inference path that treats the retrieved clip as observation evidence. The six-case qualitative pilot gives a first sanity check for this story: when the retrieved clip contains a useful trajectory, the generated video tends to exhibit more scene-level motion than the image-to-video baseline. At the same time, the pilot makes the core limitation visible. Observation-aware motion is not automatically observation-faithful motion; too much retrieved-video influence can produce blur, subject drift, or framing changes that weaken spatial consistency.

Figure 3 summarizes the reason for using the existing in-context attention update rather than explicit ETKF at video-generation scale. The Moving MNIST study shows that larger ETKF ensembles improve RMSE and PCC, but the runtime ratio grows rapidly. A DiT generator cannot practically sample and update hundreds of full video trajectories at each denoising step. In-context attention is therefore not only a modeling choice, but also a computational approximation to the forecast-observation-analysis update: it lets forecast tokens interact with observation tokens inside one denoising pass.

The main risk is also clear: the observation path must preserve temporal detail that is useful for judging motion smoothness and physics consistency. If the retrieved clip is semantically related but dynamically mismatched, the attention scores may still inject weak or distracting motion evidence.

Another limitation is that the current formulation treats attention weights as a model-internal compatibility score, but these weights are not automatically calibrated probabilities. They should be evaluated empirically through motion smoothness, physical consistency, and retrieval relevance before being interpreted as a faithful uncertainty estimate. The current qualitative comparison is still small, single-seed, and manually curated, so it should be interpreted as a pilot study rather than a clean benchmark. Its value is narrower: it shows the practical behavior of using a retrieved video as an observation in the available inference system.

7 Related Work

Trinity consistency for world models. Wei et al. [23] argue that a general world model should be defined through modal, spatial, and temporal consistency. We adopt this trinity as the conceptual frame for the paper. Modal consistency connects prompts and visual observations to a shared semantic state. Spatial consistency preserves geometry, identity, and layout across frames and viewpoints. Temporal consistency governs causal evolution and motion. Our DA framing focuses on the intersection of spatial and temporal consistency: a retrieved real-video trajectory is treated as an observation that can stabilize the DiT forecast state.

Video generation as world modeling. Latent world models compress observations into a state representation and then evolve that representation forward with a learned dynamics model. This design underlies both classical latent control architectures and modern generative world models. Ha and Schmidhuber [7] made the paradigm explicit by separating representation, memory, and control, while recent surveys argue that modern video generators are increasingly being interpreted through a world-model lens [8, 22]. Under this lens, an image-to-video DiT is not only a renderer but also

a latent transition model: it predicts how a scene state should evolve under a prompt, intent, or task description.

Spatial consistency and geometry-aware memory. A growing body of work attacks spatial inconsistency directly. World-consistent Video Diffusion uses explicit 3D coordinate frames to supervise 3D-consistent generation [27]. GEN3C conditions generation on a 3D cache to improve world-consistent video generation with precise camera control [15]. Video world models with long-term spatial memory store and retrieve geometry-grounded memory to prevent scene forgetting during revisits [25]. Other memory-based systems, including WorldPack and EvoWorld, improve spatial consistency through compressed or explicit memory mechanisms [13, 21]. These works motivate our use of one retrieved real-video trajectory as a spatial-temporal observation source.

State drift, memory, and physical plausibility. A related 2025–2026 line of work attacks temporal drift, weak state persistence, and physically implausible dynamics by strengthening memory, context, or closed-loop interaction. Examples include retrieval- or state-conditioned video world models [5, 24], embodied rollouts with mixture-of-experts or closed-loop training [17, 26], and persistent or embodiment-aware latent states [29, 6]. These works demonstrate that better state representations and memory mechanisms are critical, but they still largely handle external evidence through conditioning, retrieval augmentation, or latent persistence rather than through an explicit forecast–observation–analysis update inside the transition model.

Retrieval as external world memory. Retrieval and external memory have become common tools for improving video realism and long-horizon coherence. Retrieval-augmented generation appears in systems such as Peruzzo et al. [14] and Chen et al. [5], while explicit memory banks or persistent scene stores are central to Oshima et al. [13], Wang et al. [21], and Wu et al. [24]. In most of these methods, retrieved evidence is used as conditioning context, as a state prompt, or as an auxiliary memory. Our formulation is deliberately narrower and more explicit: retrieved evidence is treated as an observation of real-world dynamics. The retrieval memory is therefore an empirical world-memory bank, and the DiT must assimilate this observation into its latent forecast rather than merely append it as context.

Multimodal retrieval support. The formulation depends on an observation bank that is both prompt-annotated and large enough to supply semantically related real clips. OpenVid-1M is one practical choice for this role because it provides over one million text-video pairs and a curated high-definition subset [12]. For retrieval and prompt-alignment scoring we use LanguageBind, which places text, image, and video features into a shared representation space [30].

Video-as-prompt control. Video-as-Prompt (VAP) frames semantic-controlled video generation as in-context generation: a reference video provides the desired semantics, and a reference image provides the visual anchor [1]. We use this inference structure directly: the retrieved OpenVid clip becomes the reference video, and the input image remains the reference image. Our contribution is not a new attention mechanism; it is the DA interpretation and retrieval wrapper that treats the reference video as an observation trajectory from world memory and reads the forecast-to-observation token interaction as an analysis update for the DiT transition state.

Data assimilation with learned and generative priors. DA provides a principled answer to sequential state estimation under model error and partial observations. The broader DA literature has already argued that latent and reduced-order spaces are natural places to fuse machine learning with estimation [3]. More recent work combines DA with learned priors or generative models, including sparse-observation weather DA [11], derivative-free ensemble guidance for diffusion models [28], stochastic-interpolant DA [4], and diffusion or flow-based perspectives on posterior correction and sequential warm starts [9, 16, 10]. These papers show that DA-style correction remains useful even when the predictive prior is highly expressive. However, they are not focused on video world models with retrieved visual observation memory. That gap is where our method sits.

8 Conclusion

This paper reframes retrieval-guided Video-as-Prompt inference as a data-assimilation process for observation-aware video world models. The central object is the interaction between forecast tokens and the single retrieved real-video observation token sequence inside the existing DiT in-context pathway. We specify how retrieval supplies one observation trajectory from world memory, how the retrieved clip becomes a reference-video observation, and how the resulting token interaction can be read as a forecast–observation–analysis update for noise prediction and dynamics scoring. The Moving MNIST study provides an initial controlled check that this DA decomposition can improve dynamical rollout quality when observations are available. The six-case qualitative pilot then shows the practical promise and limitations of the retrieved-video path: it can increase dynamic motion, but can also introduce blur, layout drift, and extra runtime.

Future work. This draft is intended as a workshop or small-paper starting point, not a finished benchmark paper. The next steps are to run a controlled same-backbone comparison, replace the manually supplied reference clips with automatic LanguageBind retrieval from the cached OpenVid memory, evaluate more seeds and prompts, and add human or metric-based judgments of motion smoothness and physical consistency. A stronger implementation should also expose or tune the forecast-to-observation attention strength, reduce runtime, and test whether the attention update can retain the motion benefit while avoiding the spatial drift seen in the pilot.

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